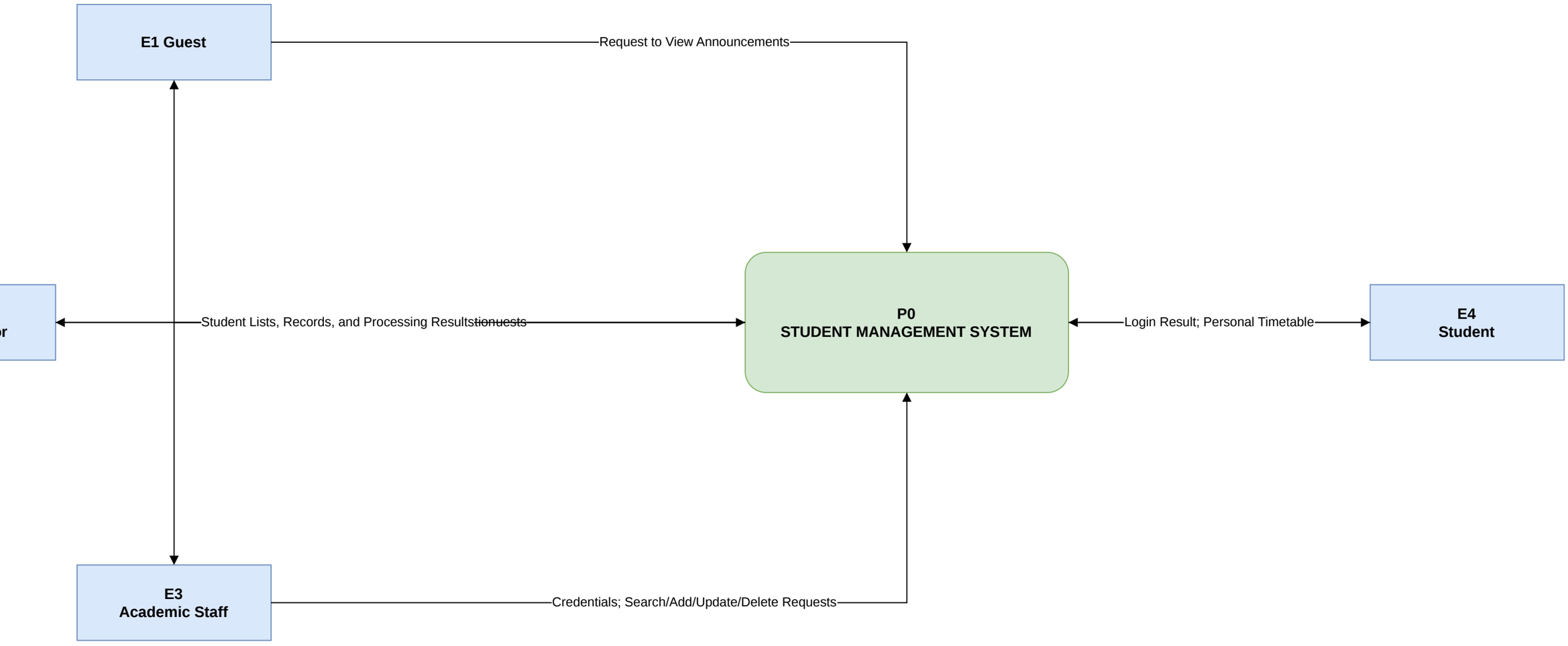
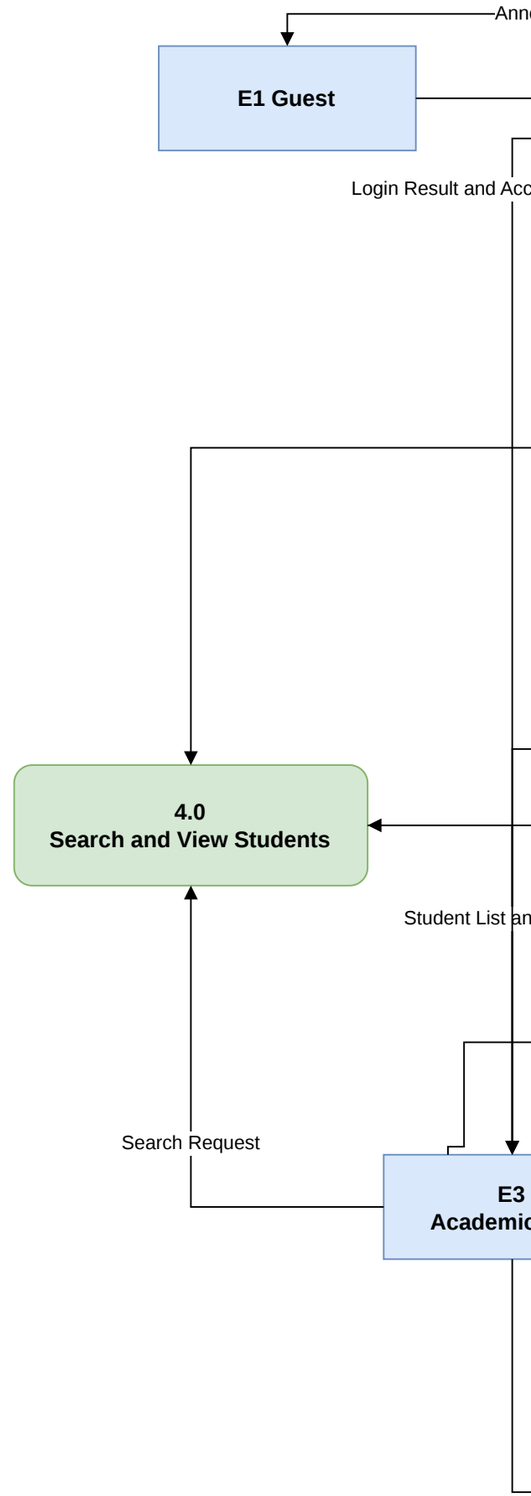
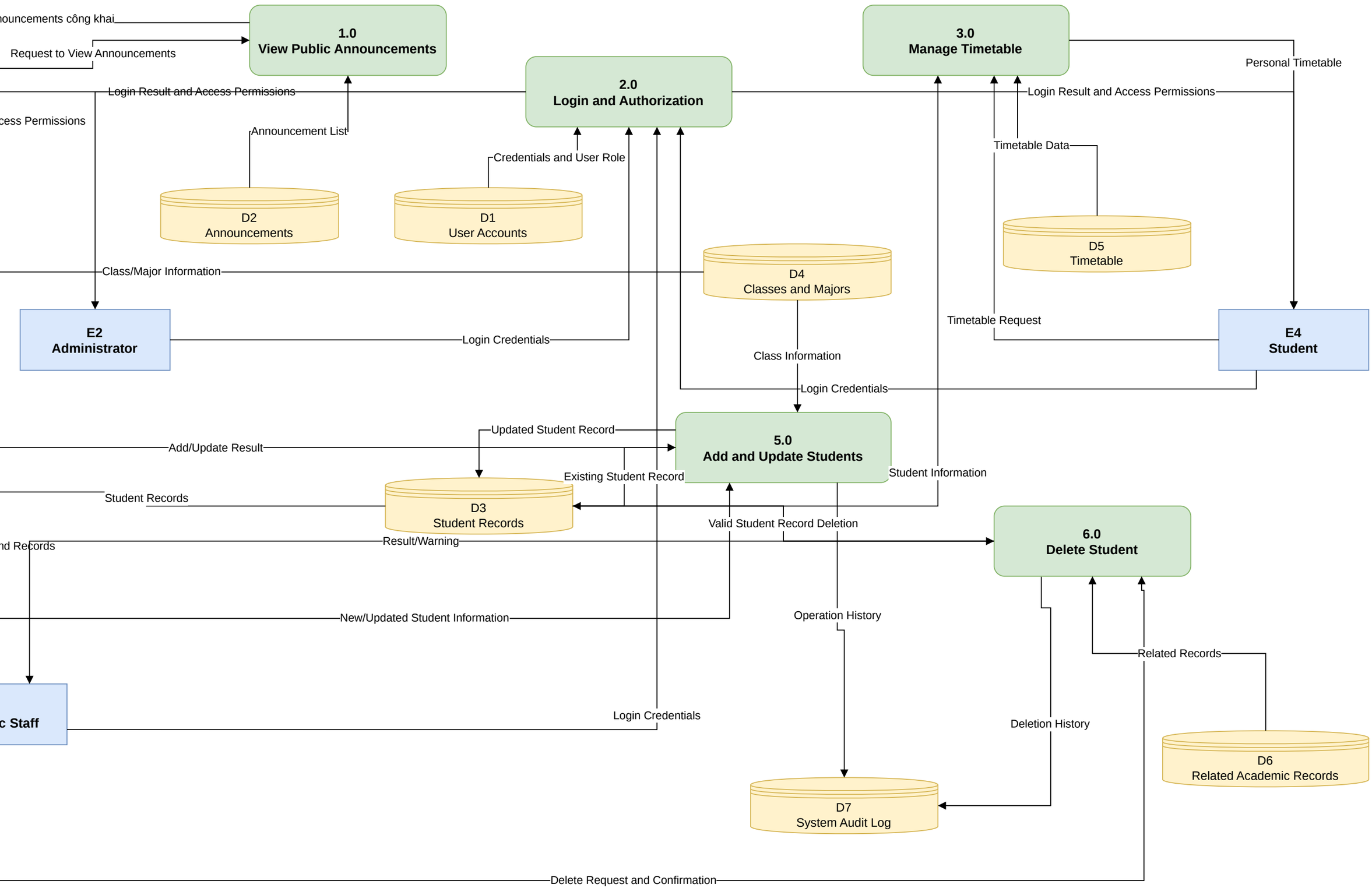
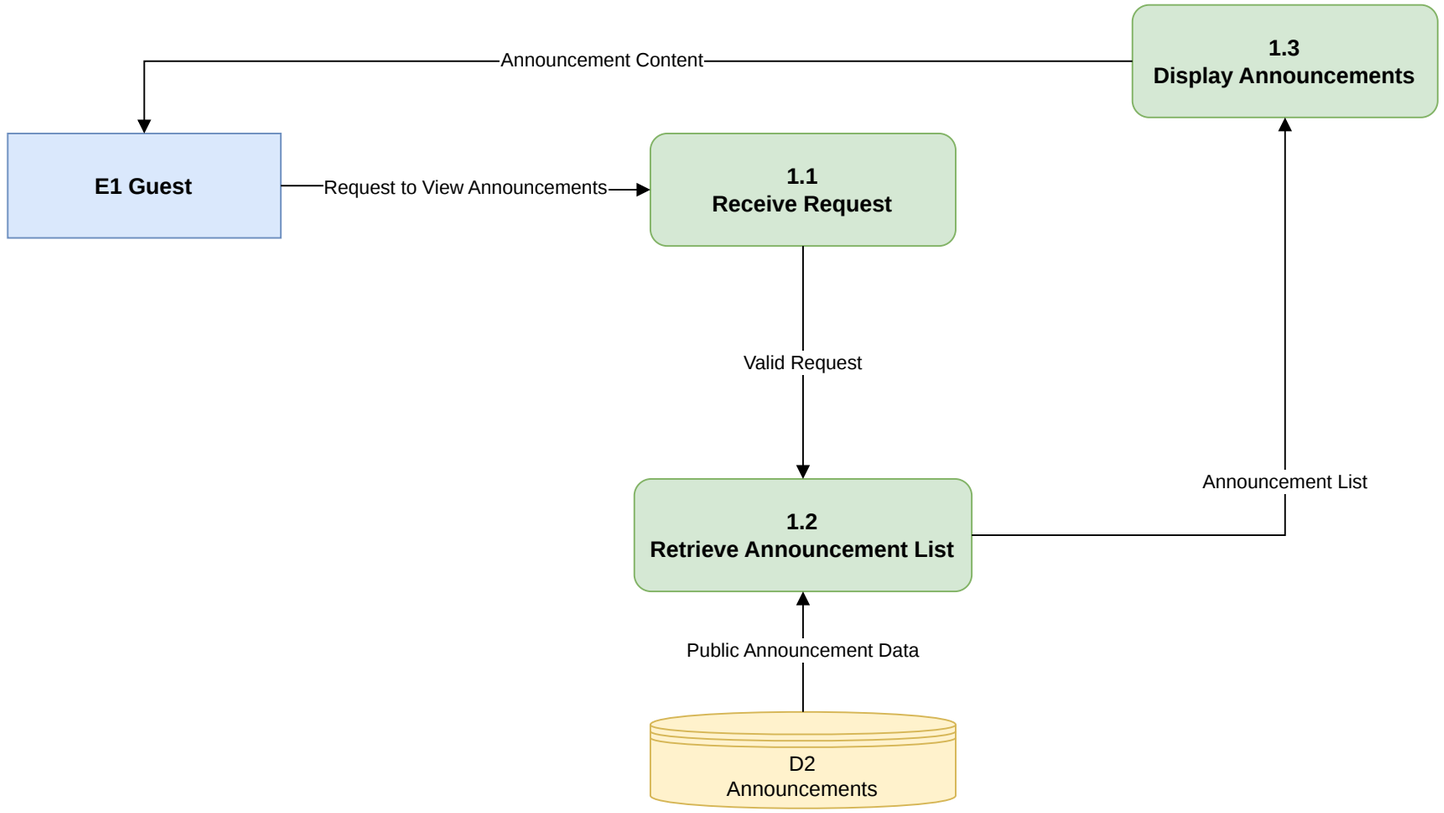
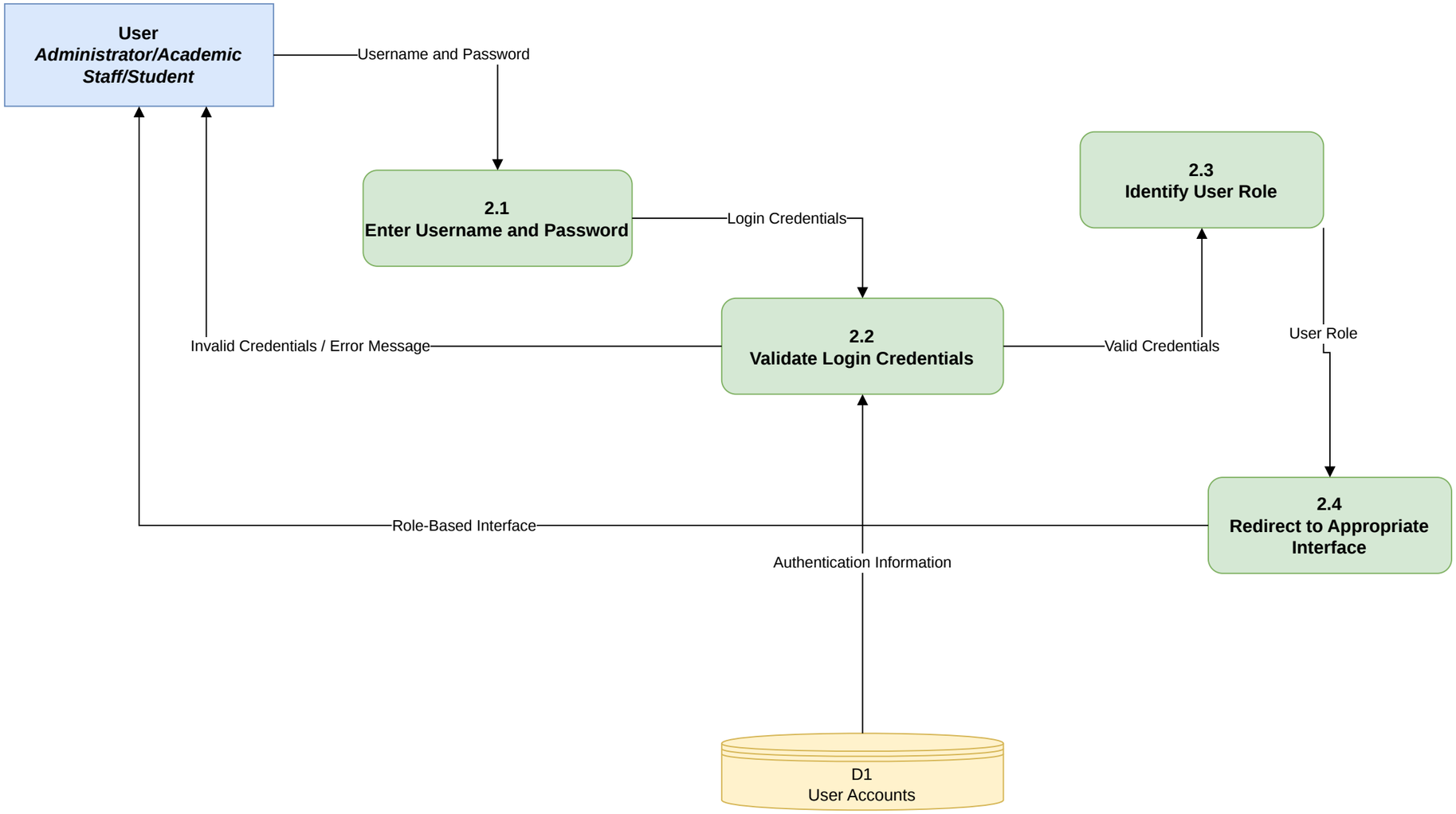
E2
Administrator

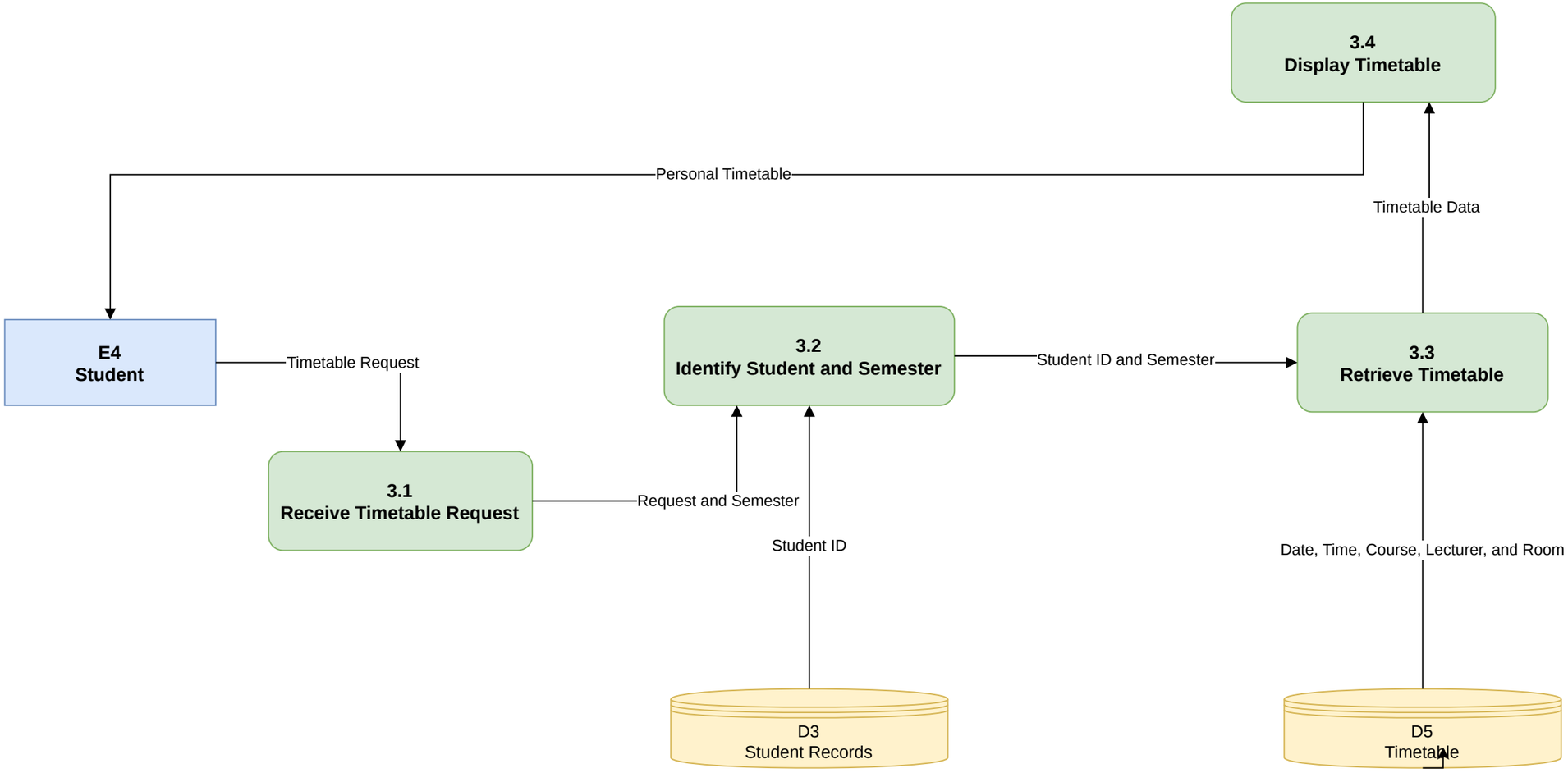

E1 Guest
Request to View Announcements
P0
STUDENT MANAGEMENT SYSTEM
E4
Student
Login Result; Personal Timetable
Student Lists, Records, and Processing Results
E3
Academic Staff
Credentials; Search/Add/Update/Delete Requests

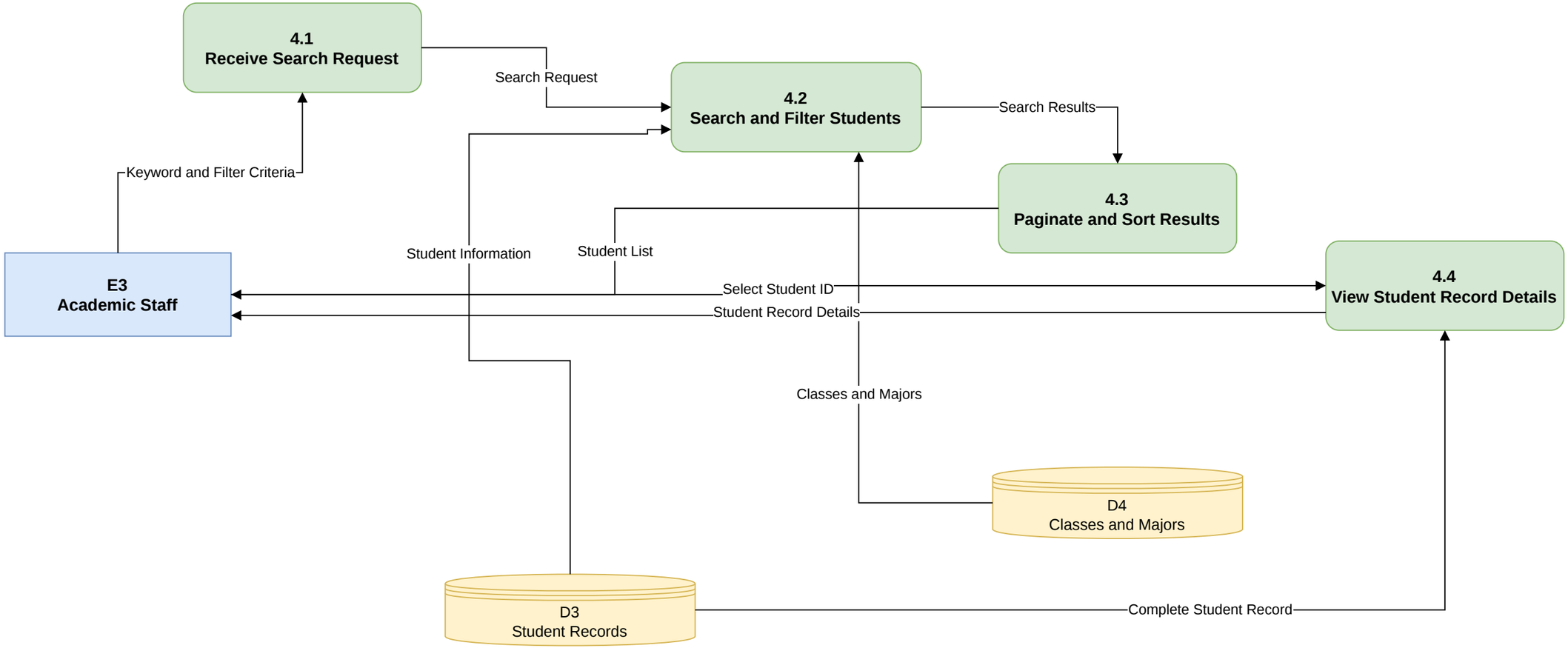

E1 Guest
Login Result and Acc
4.0
Search and View Students
Student List an
Search Request
E3
Academic

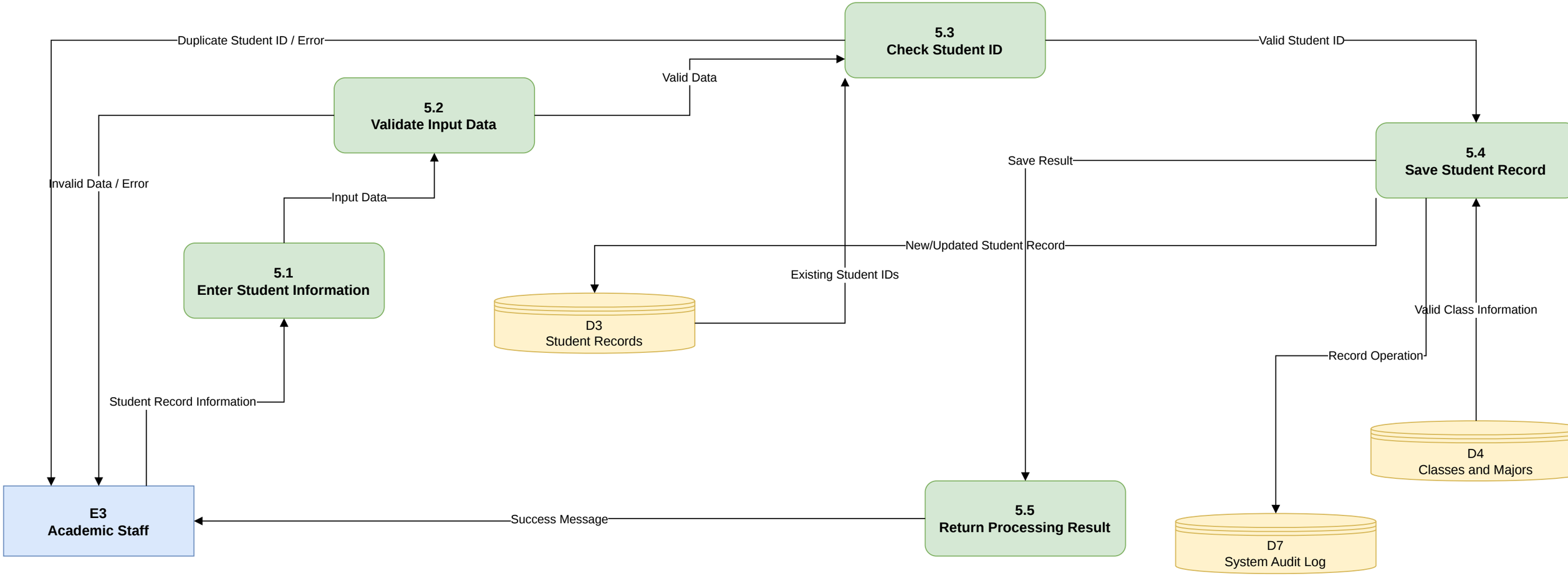

nouncements công khai
Request to View Announcements
1.0
View Public Announcements
3.0
Manage Timetable
Personal Timetable
Login Result and Access Permissions
2.0
Login and Authorization
Login Result and Access Permissions
cess Permissions
Announcement List
Credentials and User Role
Timetable Data
D2
Announcements
D1
User Accounts
D5
Timetable
Class/Major Information
D4
Classes and Majors
E2
Administrator
Login Credentials
Timetable Request
E4
Student
Class Information
Login Credentials
Updated Student Record
5.0
Add and Update Students
Add/Update Result
Existing Student Record
Student Information
Student Records
D3
Student Records
Valid Student Record Deletion
6.0
Delete Student
nd Records
Result/Warning
New/Updated Student Information
Operation History
Related Records
c Staff
Login Credentials
Deletion History
D6
Related Academic Records
D7
System Audit Log
Delete Request and Confirmation


1.3
Display Announcements
Announcement Content
E1 Guest
Request to View Announcements
1.1
Receive Request
Valid Request
Announcement List
1.2
Retrieve Announcement List
Public Announcement Data
D2
Announcements

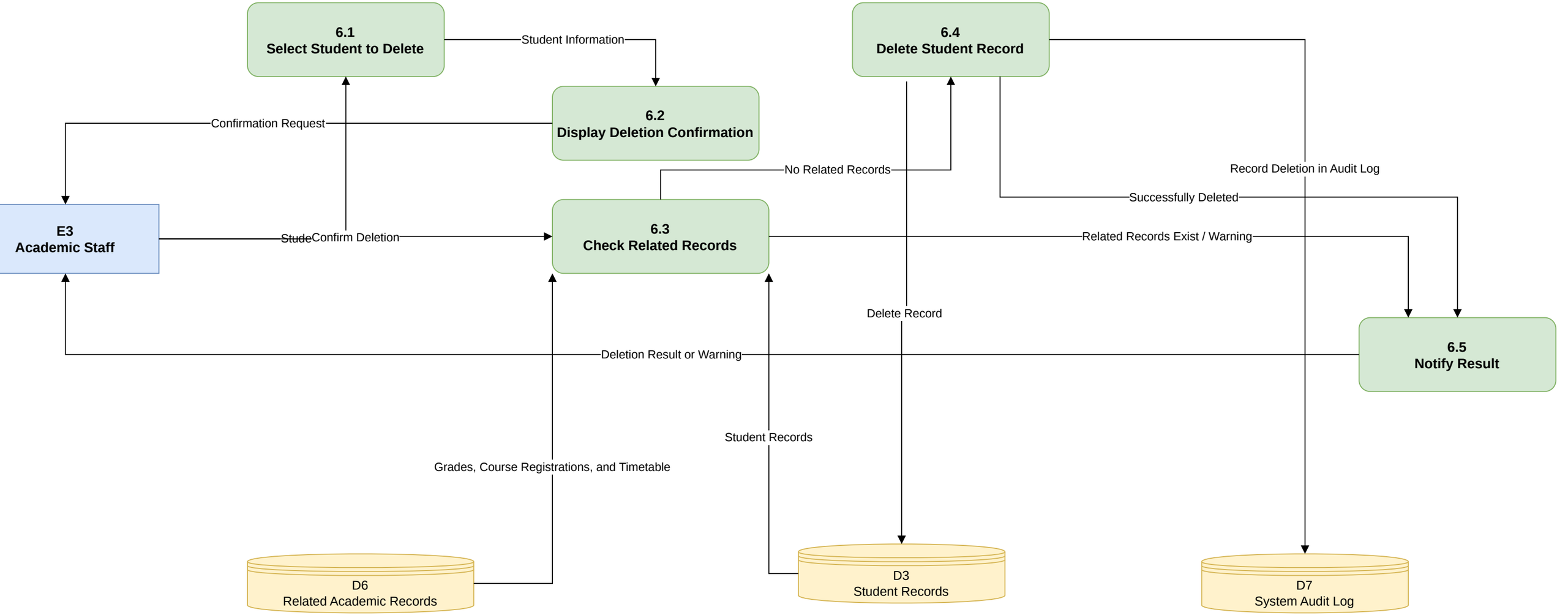

User
Administrator/Academic Staff/Student
Username and Password
2.1
Enter Username and Password
Login Credentials
2.3
Identify User Role
Invalid Credentials / Error Message
2.2
Validate Login Credentials
Valid Credentials
User Role
Role-Based Interface
2.4
Redirect to Appropriate Interface
Authentication Information
D1
User Accounts

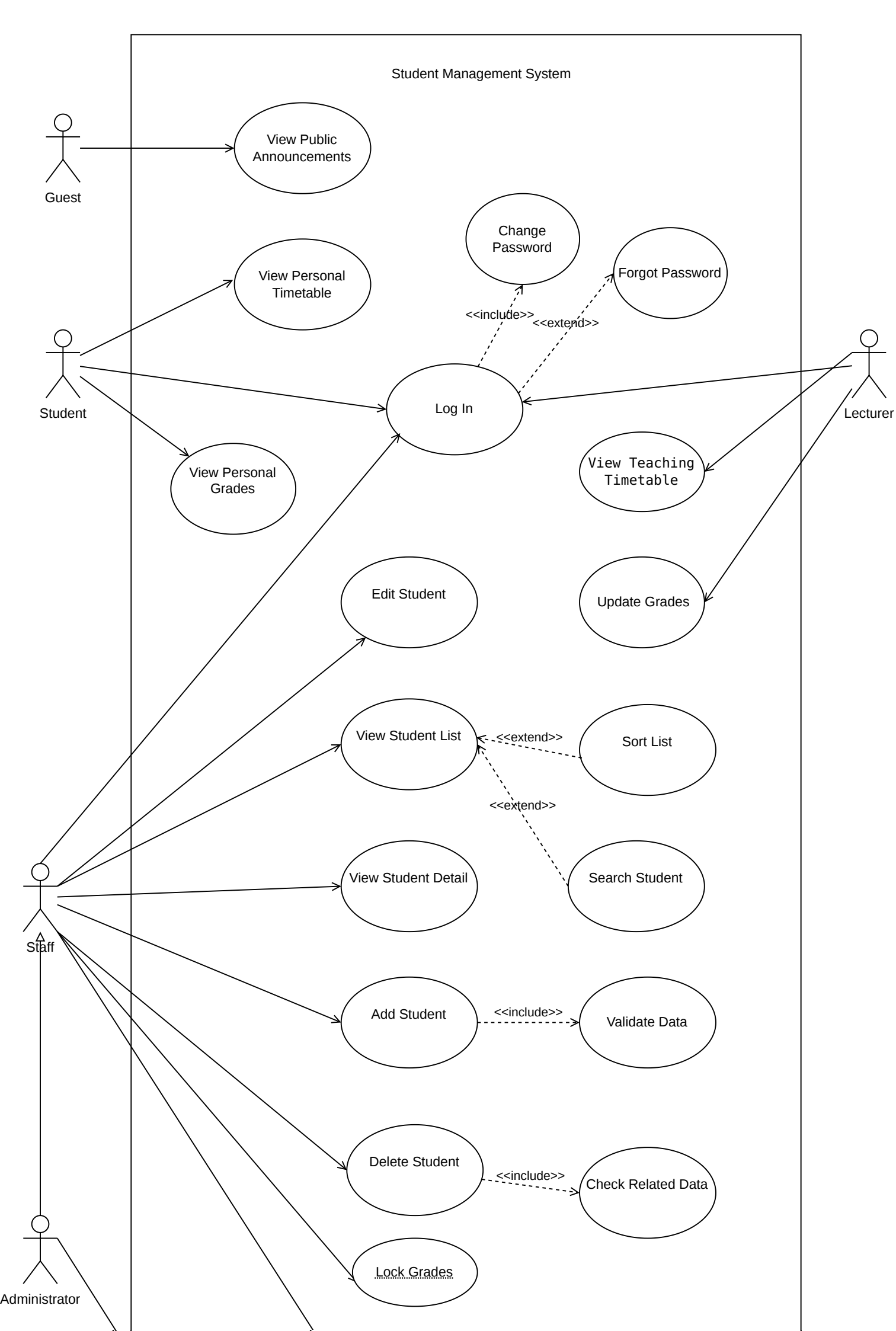

3.4
Display Timetable
Personal Timetable
Timetable Data
E4
Student
Timetable Request
3.2
Identify Student and Semester
Student ID and Semester
3.3
Retrieve Timetable
3.1
Receive Timetable Request
Request and Semester
Student ID
Date, Time, Course, Lecturer, and Room
D3
Student Records
D5
Timetable

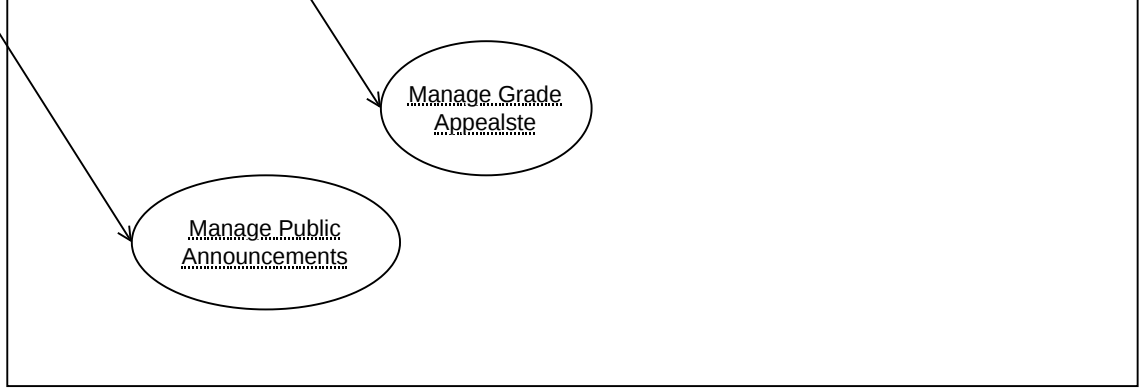

4.1
Receive Search Request
Search Request
4.2
Search and Filter Students
Search Results
4.3
Paginate and Sort Results
Keyword and Filter Criteria
E3
Academic Staff
Student Information
Student List
Select Student ID
Student Record Details
4.4
View Student Record Details
Classes and Majors
D4
Classes and Majors
D3
Student Records
Complete Student Record

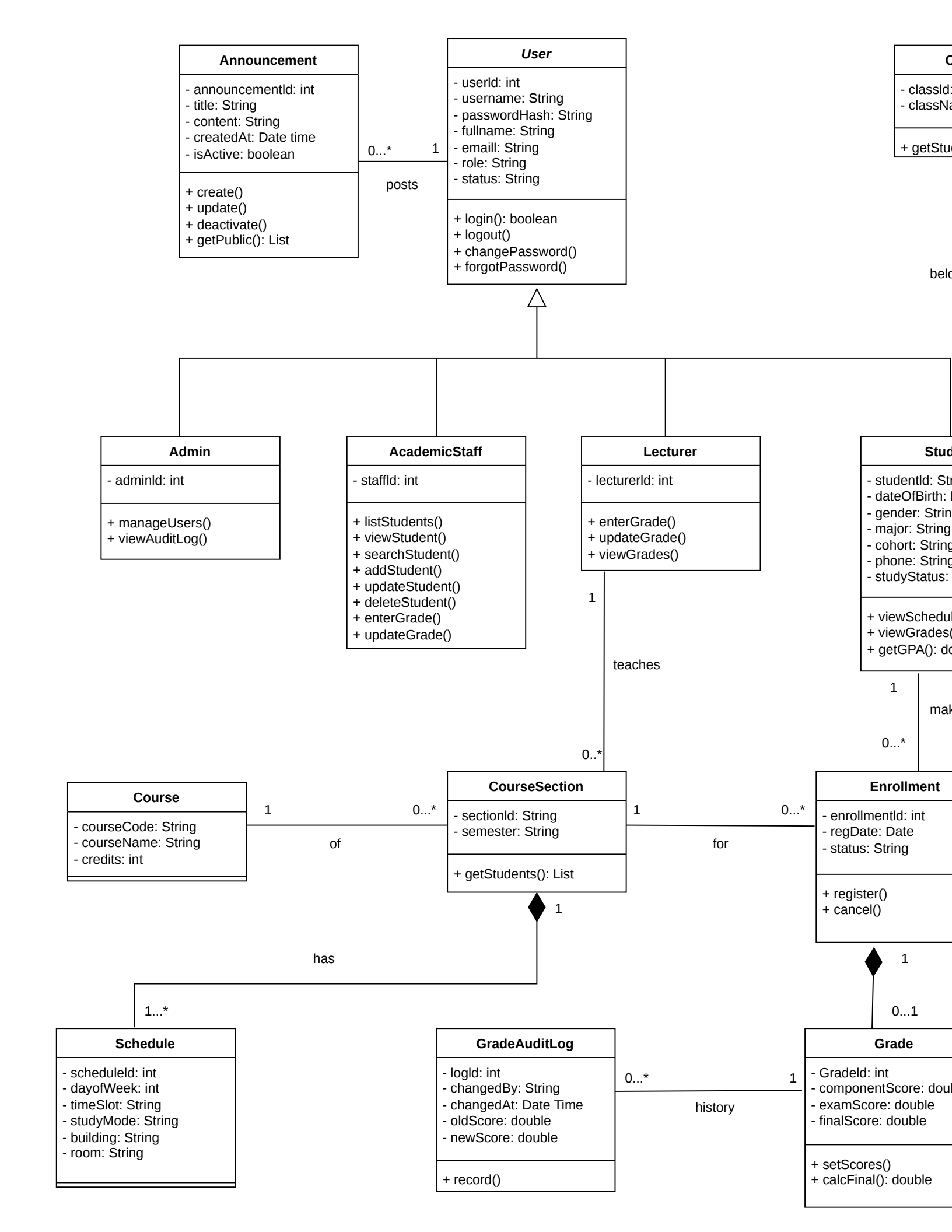

5.3
Check Student ID
Duplicate Student ID / Error
Valid Student ID
Valid Data
5.2
Validate Input Data
5.4
Save Student Record
Save Result
Invalid Data / Error
Input Data
5.1
Enter Student Information
New/Updated Student Record
Existing Student IDs
D3
Student Records
Valid Class Information
Record Operation
D4
Classes and Majors
Student Record Information
E3
Academic Staff
Success Message
5.5
Return Processing Result
D7
System Audit Log

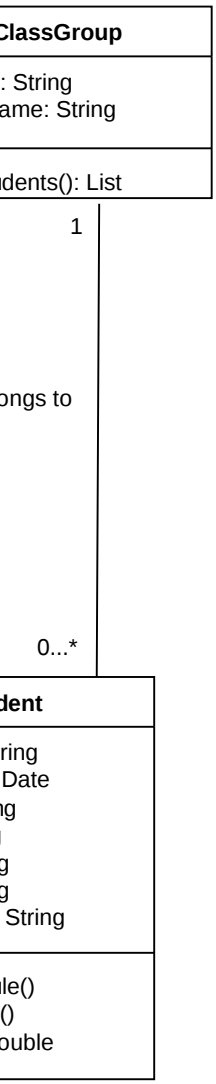



6.1
Select Student to Delete
Student Information
6.2
Display Deletion Confirmation
Confirmation Request
E3
Academic Staff
Stude
Confirm Deletion
6.3
Check Related Records
No Related Records
6.4
Delete Student Record
Record Deletion in Audit Log
Successfully Deleted
Related Records Exist / Warning
6.5
Notify Result
Deletion Result or Warning
Delete Record
Student Records
Grades, Course Registrations, and Timetable
D6
Related Academic Records
D3
Student Records
D7
System Audit Log

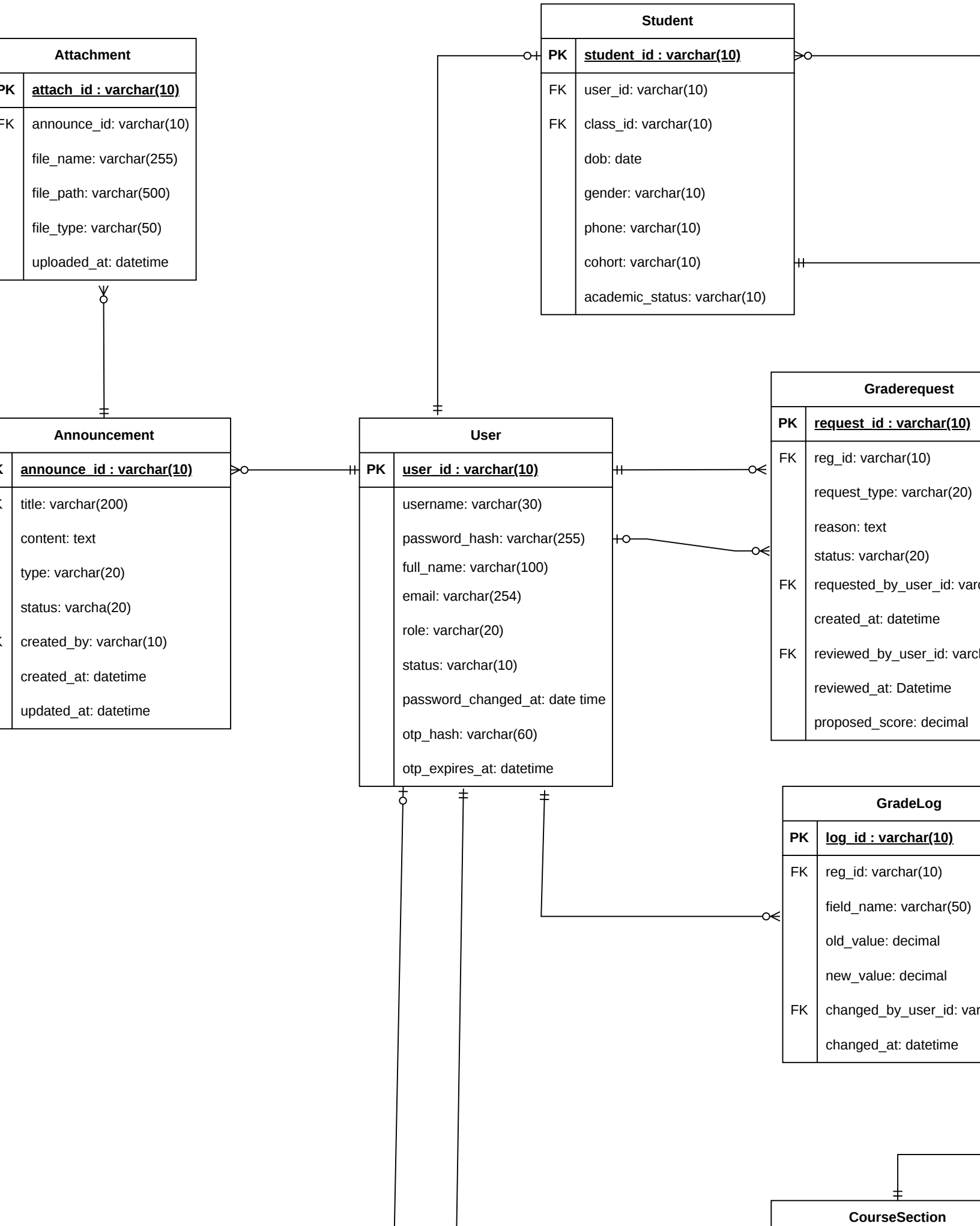

Student Management System
Guest
View Public Announcements
Change Password
Forgot Password
View Personal Timetable
<<include>>
<<extend>>
Student
Log In
Lecturer
View Personal Grades
View Teaching Timetable
Edit Student
Update Grades
View Student List
<<extend>>
Sort List
<<extend>>
View Student Detail
Search Student
Staff
Add Student
<<include>>
Validate Data
Delete Student
<<include>>
Check Related Data
Lock Grades
Administrator

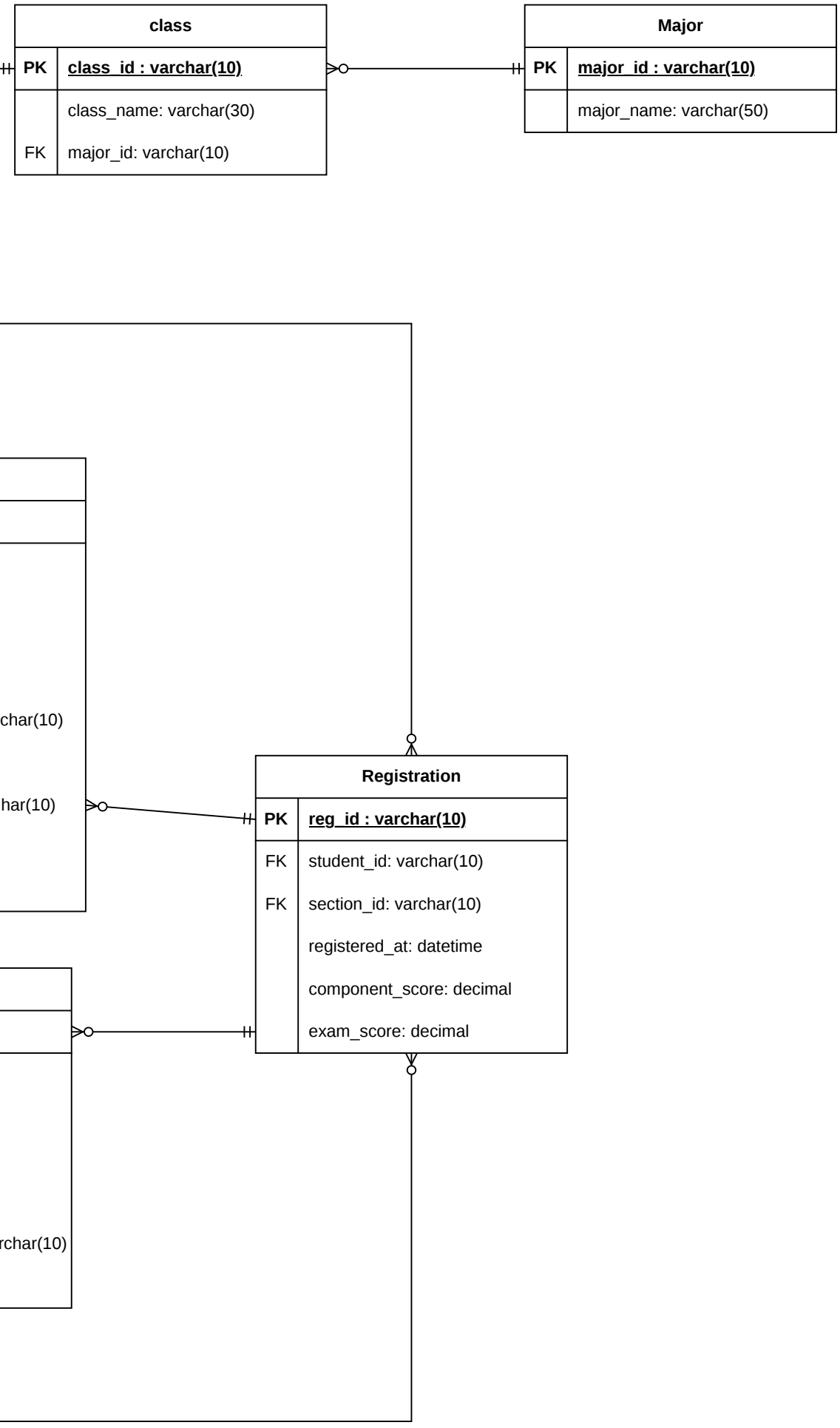

Manage Grade Appealste
Manage Public Announcements

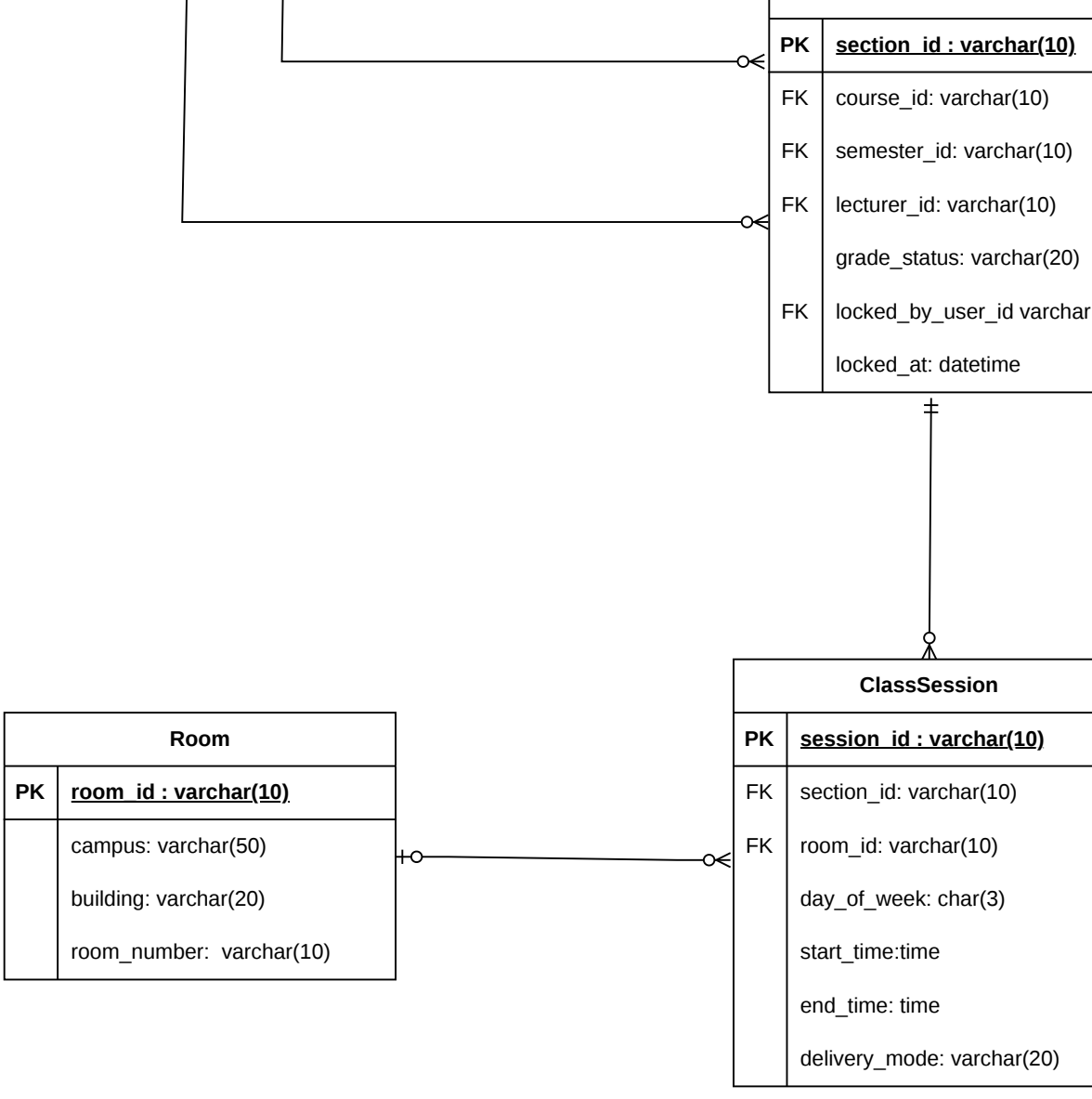

Announcement
- announcementId: int
- title: String
- content: String
- createdAt: Date time
- isActive: boolean
+ create()
+ update()
+ deactivate()
+ getPublic(): List
0...*
1
posts
User
- userId: int
- username: String
- passwordHash: String
- fullname: String
- emaill: String
- role: String
- status: String
+ login(): boolean
+ logout()
+ changePassword()
+ forgotPassword()
Admin
- adminId: int
+ manageUsers()
+ viewAuditLog()
AcademicStaff
- staffId: int
+ listStudents()
+ viewStudent()
+ searchStudent()
+ addStudent()
+ updateStudent()
+ deleteStudent()
+ enterGrade()
+ updateGrade()
Lecturer
- lecturerId: int
+ enterGrade()
+ updateGrade()
+ viewGrades()
1
teaches
0..*
Course
- courseCode: String
- courseName: String
- credits: int
1
0...*
of
CourseSection
- sectionId: String
- semester: String
+ getStudents(): List
1
0...*
for
Enrollment
- enrollmentId: int
- regDate: Date
- status: String
+ register()
+ cancel()
1
has
1...*
1
0...1
Schedule
- scheduleId: int
- dayofWeek: int
- timeSlot: String
- studyMode: String
- building: String
- room: String
GradeAuditLog
- logId: int
- changedBy: String
- changedAt: Date Time
- oldScore: double
- newScore: double
+ record()
0...*
1
history
Grade
- GradeId: int
- componentScore: doub
- examScore: double
- finalScore: double
+ setScores()
+ calcFinal(): double

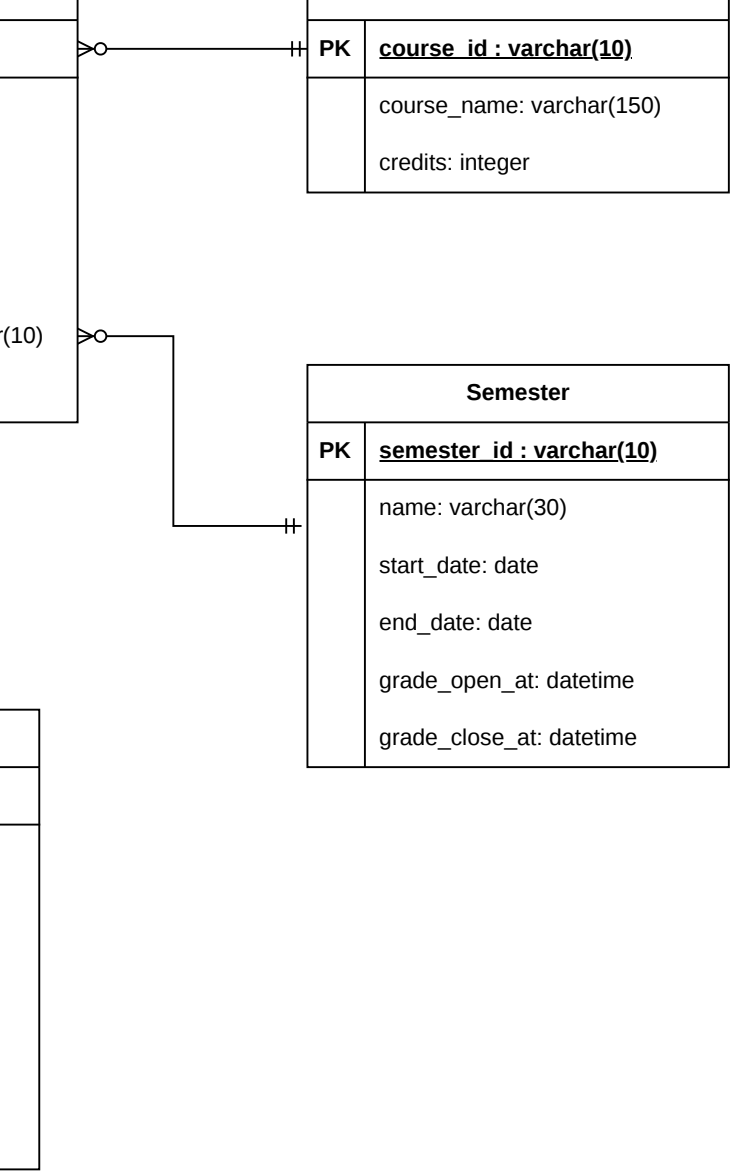

ClassGroup
: String
ame: String
dents(): List
1
ongs to
0...*
dent
ring
Date
g
g
g
String
le()
()
ouble
kes
ble




Attachment
PK
attach_id : varchar(10)
FK
announce_id: varchar(10)
file_name: varchar(255)
file_path: varchar(500)
file_type: varchar(50)
uploaded_at: datetime
Student
PK
student_id : varchar(10)
FK
user_id: varchar(10)
FK
class_id: varchar(10)
dob: date
gender: varchar(10)
phone: varchar(10)
cohort: varchar(10)
academic_status: varchar(10)
Announcement
announce_id : varchar(10)
title: varchar(200)
content: text
type: varchar(20)
status: varcha(20)
created_by: varchar(10)
created_at: datetime
updated_at: datetime
User
PK
user_id : varchar(10)
username: varchar(30)
password_hash: varchar(255)
full_name: varchar(100)
email: varchar(254)
role: varchar(20)
status: varchar(10)
password_changed_at: date time
otp_hash: varchar(60)
otp_expires_at: datetime
Graderequest
PK
request_id : varchar(10)
FK
reg_id: varchar(10)
request_type: varchar(20)
reason: text
status: varchar(20)
FK
requested_by_user_id: varc
created_at: datetime
FK
reviewed_by_user_id: varch
reviewed_at: Datetime
proposed_score: decimal
GradeLog
PK
log_id : varchar(10)
FK
reg_id: varchar(10)
field_name: varchar(50)
old_value: decimal
new_value: decimal
FK
changed_by_user_id: var
changed_at: datetime
CourseSection


class
PK
class_id : varchar(10)
class_name: varchar(30)
FK
major_id: varchar(10)
Major
PK
major_id : varchar(10)
major_name: varchar(50)
char(10)
har(10)
Registration
PK
reg_id : varchar(10)
FK
student_id: varchar(10)
FK
section_id: varchar(10)
registered_at: datetime
component_score: decimal
exam_score: decimal
rchar(10)
Course


PK
section_id : varchar(10)
FK
course_id: varchar(10)
FK
semester_id: varchar(10)
FK
lecturer_id: varchar(10)
grade_status: varchar(20)
FK
locked_by_user_id varchar
locked_at: datetime
ClassSession
PK
session_id : varchar(10)
FK
section_id: varchar(10)
FK
room_id: varchar(10)
day_of_week: char(3)
start_time:time
end_time: time
delivery_mode: varchar(20)
Room
PK
room_id : varchar(10)
campus: varchar(50)
building: varchar(20)
room_number: varchar(10)


PK
course_id : varchar(10)
course_name: varchar(150)
credits: integer
(10)
Semester
PK
semester_id : varchar(10)
name: varchar(30)
start_date: date
end_date: date
grade_open_at: datetime
grade_close_at: datetime